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Post-Oligocene tectonic rotation of the Oregon Western Cascade Range and the Klamath Mountains

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ABSTRACT

Paleomagnetic data indicate that the Western Cascade Range from northern California to central Washington has undergone about 27° of clockwise rotation since 25 m.y. B.P. All the available regional paleomagnetic data fit a two-phase rotation model, the first phase being an Eocene rotation that accompanied the accretion of the Oregon Coast Range to the continent and the second phase being a rotation associated with post-early Miocene extension in the Basin and Range province.

INTRODUCTION

Anomalous directions of magnetization in Tertiary rocks of the Coast Range of Oregon (Fig. 1; Simpson and Cox, 1977; Beck and Plumley, 1980; Magill and others, 1981) have established that Cenozoic sedimentary and volcanic rocks have undergone clockwise rotations about vertical axes of as much as 75°. The observation that similar clockwise rotations are recorded in rocks of varying Cenozoic age as petrologically dissimilar as basalts and turbidites leaves little doubt that the anomalous paleomagnetic directions have recorded true tectonic rotations. The paleomagnetic data and regional geology suggest that the Coast Range rotated as a single coherent block extending from the Klamath Mountains northward to the Oregon-Washington border. The geologic observations that support this interpretation include the apparent absence of major crosscutting tectonic structures, the similarity of rotations found throughout the block, the continuity of formation boundaries, and the consistency of current directions within the middle Eocene sediments that extend the length of the Coast Range (Snavely and others, 1964; Simpson and Cox, 1977; Magill and others, 1981).

In attempting to understand the paleomagnetically observed rotation of the Coast Range in the context of the geologic history of the region, it is important to determine the regional extent of the rotated block and the timing of rotation as precisely as possible. Post-Oligocene rocks of the Coast Range (Clark, 1969; Beck and Plumley, 1980) and of the Western Cascades of Washington (Bates and others, 1979; R. G. Bates and M. E. Beck, Jr., in prep.; Beck and Burr, 1979) are rotated significantly less than are Eocene rocks (Simpson and Cox, 1977; Magill and others, 1981), implying that significant rotation occurred before as well as after Oligocene time. Our new paleomagnetic results from upper Oligocene and lower Miocene Cascade rocks help to establish and refine these constraints on the timing and spatial extent of the rotation.

Several lines of evidence suggest that the coherently rotated Coast Range block extends north to the Washington border but does not include the Coast Range of Washington. Gravity anomalies over the Oregon Coast Range are smooth and of long wavelength, suggesting a deep, elongate regional structure, whereas in Washington short-wavelength anomalies with sharp gradients are suggestive of shallow, local, crustal tectonic structures. However, local near-surface density contrasts due to igneous and erosional processes cannot be dismissed as possible causes of some of these anomalies. North of the Oregon-Washington border, paleomagnetically recorded rotations vary from region to region and are somewhat smaller than the rotations recorded in rocks of similar age to the south (Globerman and Beck, 1979; Wells and Coe, 1979). The emerging picture is one of different amounts of rotation in different tectonic blocks in the Washington Coast Range contrasted to the coherent rotation of one Coast Range block to the south.

In southern Oregon the dominantly Cenozoic terrane of the Coast Range

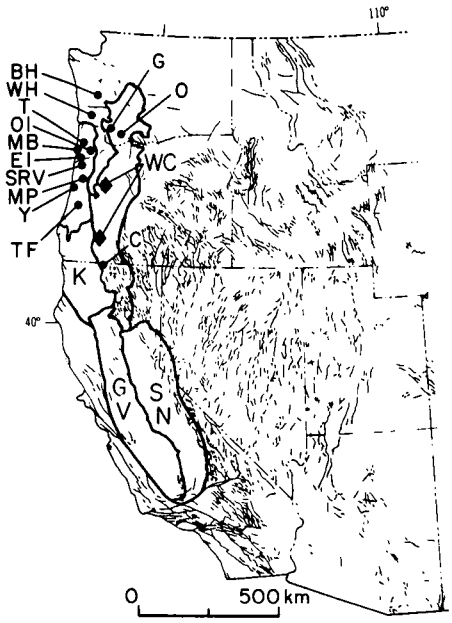


Figure 1. Generalized tectonic map of western North America showing steeply dipping faults, our proposed tectonic blocks, and location of paleomagnetic studies in Washington and Oregon Coast Range and Cascades. Longer faults are generally strike-slip faults; shorter ones are normal faults. Map shows faults active in past 10 to 15 m.y. for which some Quaternary movement is suspected. Tectonic blocks (heavy line boundaries): K = Klamath Mountains, SN = Sierra Nevada, GV = Great Valley. Diamonds = sites where paleomagnetic results from Western Cascades described in this paper (WC) were collected. Other paleomagnetic studies in Cascade block (large dots): G = Goble Volcanic Series (Beck and Burr, 1979); O = Ohanapecosh Formation (R. G. Bates and M. E. Beck, Jr., in prep.); C = California Cascades (Beck, 1962). Paleomagnetic studies in Oregon Coast Range tectonic block (large dots): T = Tillamook Volcanic Series (Magill and others, 1981); MB = Miocene basalts; SRV = Siletz River Volcanics; Y = Yachats basalts; TF = Tyee and Flournoy Formations (Simpson and Cox, 1977); EI = Eocene intrusions; OI = Oligocene intrusions (Beck and Plumley, 1980); MP = Mary's Peak (Clark, 1969). Paleomagnetic sampling in Washington Coast Range is limited to BH = Black Hills (Globerman and Beck, 1979) and WH = Willipa Hills (Wells and Coe, 1979). (Faults are from Eaton, 1980.)

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adjoins the mainly Mesozoic terrane of the Klamath Mountains (Fig. 1) along a complex geologic boundary. Generally, the sediments of the Coast Range appear to be in depositional contact with the older rocks of the Klamath Mountains, but moderate folding and faulting complicate the depositional contact and prevent an unambiguous conclusion as to whether or not the Klamath terrane rotated with the Oregon Coast Range during the Cenozoic. This tectonic relationship has not been tested paleomagnetically. The lower Tertiary rocks of the Coast Range pass to the east and northeast beneath the middle and upper Tertiary and Quaternary volcanic and sedimentary rocks of the Cascade volcanic chain. The question of whether the Cascade Range has participated in the rotation of the Coast Range is the subject of this paper and of other current research (Bates and others, 1979; Beck and Burr, 1979; R. G. Bates and M. E. Beck, Jr., in prep.).

WESTERN CASCADE RANGE

Geology and Paleomagnetic Data

The Cascade Range extends nearly due north from northern California into British Columbia. The Pliocene and Quaternary shield volcanoes, stratovolcanoes, and cinder cones of the High Cascades are underlain by upper Eocene to Miocene basalts, basaltic andesites, and andesite flows and pyroclastic rocks (Peck and others, 1964; Wise, 1969) which crop out primarily on the western side of the High Cascades and form the Western Cascade Range.

Paleomagnetic samples were collected from two regions in the Western Cascades of Oregon about 200 km apart (Fig. 1), the northern one in the Sweet Home quadrangle and the southern one in the Butte Falls quadrangle (diamond symbols in Fig. 1). The geology of the northern sampled section (Peck and others, 1964) consists of subaerial flows of basalt and basaltic andesite of the Little Butte and Sardine Formations. Geochemical and K-Ar age data show that these rocks resemble island-arc tholeiites and range in age from 20 to 30 m.y. (White and McBirney, 1979; C. M. White, 1980, oral commun.). Most of our paleomagnetic samples were taken from the flows sampled by White and McBirney, to assure good age control. The regional structure is relatively uncomplicated and has not been affected by major faulting. Regional tilts in the 700 m of sampled section are less than 5° and appear randomly oriented.

The southern sampled section has been mapped by Wilkinson and others (1941) and J. G. Smith of the U.S. Geological Survey (1980, oral commun.) as consisting of subaerial flows of basalt, basaltic andesite, ash-flow tuff, and tuff breccia within the Little Butte volcanic series. On the basis of K-Ar age determinations by J. G. Smith (1980, oral commun.), the 3,000 m of sampled section spans the time interval 20 to 30 m.y. B.P. Geochemically the rocks are calc-alkalic and are indistinguishable from volcanic rocks of the High Cascades of the same latitude (J. G. Smith, 1980, oral commun.). The rocks have a uniform regional dip of 7° to the northeast. Detailed mapping shows no faults with large displacement (J. G. Smith, 1980, oral commun.).

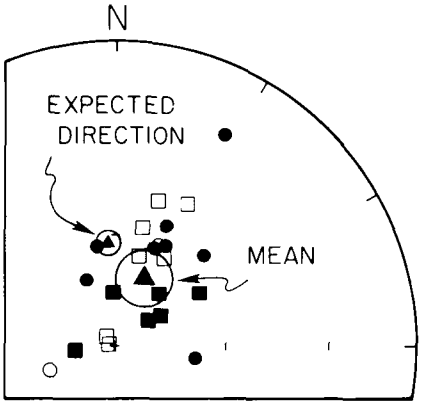


Figure 2. Paleomagnetic field directions for Western Cascade rocks corrected for tectonic tilt. Solid symbols = normal polarity; open symbols = reversed polarity. Directions of reversed polarity have been reflected through origin to plot on lower hemisphere of this stereographic polar plot. Boxes = northern sampling locality; solid circles = southern locality. Large circles about expected and observed means are 95% confidence circles ($\alpha 95$).

From five to eight samples were collected from each of 17 sites in the northern locality and 11 sites in the southern locality. All samples were magnetically cleaned in a stepwise progression, with alternating peak fields ranging from 10 to 100 mT. The directional stability and coercivity spectrum of each sample were used to determine the optimum field for cleaning. Rocks from four sites (three from the northern and one from the southern locality) were found to be magnetically unstable or to be overprinted by secondary magnetizations; these sites were not included in our final synthesis. The remaining sites were of high magnetic quality, with 9 of reversed polarity and 15 of normal polarity. Circles of 95% confidence were usually about 4° and in all cases less than 12°.

Figure 2 summarizes the site mean directions after magnetic cleaning and correction for regional tilt. Only the southern sites were corrected for tilt, because the regional character of the southern tilts suggests a tectonic origin whereas the small random tilts of the northern sites likely are original dips of the lava flows. The mean measured inclination (70°) and the expected inclination (62°) differ by 8° ± 8° with 95% confidence (see Magill and Cox, 1980). The mean declination shows a statistically significant clockwise rotation from the expected late Oligocene field direction for stable North America. Taking into account the uncertainty in the measured and expected field directions, the clockwise rotation is 25° ± 20° with 95% confidence (Table 1).

Interpretation

These data, when combined with three other Cascade paleomagnetic studies,

TABLE 1. PALEOMAGNETIC RESULTS, WESTERN CASCADES

	Pole					Field direction		Flattening		Rotation	
	Lat (°N)	Long (°E)	α_{95} (°)	N	κ	Inclination (°)	Declination (°)	F (°)	ΔF (°)	R (°)	ΔR (°)
Observed	72.2	280.0	10.9	24	8.3	70.1	20.7	-8.0	8.1	24.7	19.7
Expected	87.0	162.0	4.0	21	66.0	62.1	356.0				

Note: Expected field from Irving (1979) for 25 m.y.B.P., α_{95} = radius of 95% confidence about the mean pole, N = number of sites, κ = precision parameter, $F = \text{Inc.}_{\text{expected}} - \text{Inc.}_{\text{observed}}$, $\Delta F = [\Delta I^2_{\text{obs}} + \Delta I^2_{\text{exp}}]^{\frac{1}{2}}$ where $\Delta I = 2\alpha_{95}/1 + 3\cos^2 p$, $R = \text{Dec.}_{\text{observed}} - \text{Dec.}_{\text{expected}}$, $\Delta R = [\Delta^2_{\text{obs}} + \Delta^2_{\text{exp}}]^{\frac{1}{2}}$ where $\Delta = \sin^{-1}[\sin\alpha_{95}/\sin p]$, p = colatitude.

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strongly suggest a uniform rotation of the Cascade Range from northern California to central Washington. Paleomagnetic results from basalts and andesites of the early Cascade Goble Volcanics (32 to 45 m.y. old) and Ohanapecosh Formation (31 to 37 m.y. old) of southern Washington (Fig. 1) show rotations of 28° ± 12° (Beck and Burr, 1979) and 35° ± 12° (R. G. Bates and M. E. Beck, Jr., in prep.), respectively. Our reinterpretation of the paleomagnetic study of Beck (1962) of Western Cascade rocks (uppermost Eocene to Miocene) in northern California (Fig. 1) indicates 18° ± 14° clockwise rotation. In this early study, Beck (1962) interpreted the rotated magnetization directions as the result of improper statistical sampling or anomalous behavior of the geomagnetic field. Considered in the context of the present results from Cascade rocks of similar age, Beck's findings appear to reflect a true tectonic rotation. These three studies, combined with our results, yield an average clockwise rotation (weighted by 95% confidence limits) of 27° ± 7°. This is our best estimate of the rotation of the Cascade block. The internal consistency is impressive and is indicative of a tectonic rotation affecting the entire Western Cascades.

Rotation of the Cascade Range without concurrent rotation of the adjacent Coast Range and Klamath Mountains would require the existence of major thrust faults and shear zones between the Cascades and neighboring terranes. However, the geologic contacts of the Cascades with the adjoining Coast Range and Klamaths to the west are mostly depositional and not faulted. In south-central Oregon and northern California, upper Eocene Cascade volcanic and marine sedimentary rocks (Colestin, Spencer, and Fisher Formations) lie both conformably and unconformably on lower and middle Eocene Coast Range rocks (Umpqua and Tyee) (Wilkinson and others, 1941; Hoover, 1963) and on pre-Tertiary rocks of the Klamath Mountains (Wilkinson and others, 1941; Williams, 1949; Wells and Peck, 1961). The Cascade rocks also rest depositionally on Coast Range rocks in southern Washington (Snavely and others, 1958; Roberts, 1958; Hammond, 1979). These stratigraphic relationships and other similar depositional relationships of the paleomagnetically sampled units upon the basal formations of the Cascades require that the Coast Range and Klamaths rotated with the Cascades. We emphasize that rotation of the Klamaths is inferred from the stratigraphic contacts and has not yet been tested directly by a paleomagnetic study of Klamath rocks.

Several geologic observations suggest that in the past 25 m.y., rotation of the Cascades, Coast Range, and Klamaths resulted from internal deformation of North America and was not part of an accretionary process. The Western Cascades are located 200 to 250 km inland from the offshore trench associated with the arc volcanism of the Cascades. Because of their inland position, it seems unlikely that rotation of the Cascades was related to accretionary processes at the convergent margin. The deformation accompanying accretion is typically limited to the continental slope, shelf, and shoreline regions and is associated with subduction of the downgoing slab. Because the Coast Range and Klamaths do not show evidence of the post–25 m.y. B.P. penetrative faulting and folding that characterize the continental slope and shelf, such deformation does not appear to reach as far east as the Cascades. Late Cenozoic rotation of the Cascades, Klamaths, and Coast Range therefore appears not to have been directly related to subduction but more likely was the result of broad internal deformation within the western Cordillera.

The paleomagnetic rotation data from the Coast Range and Cascades are plotted as a function of time in Figure 3. It might appear that the best fit to the rotation data is a straight line corresponding to rotation at a constant rate. However, other curves are more consistent with the data, as may be seen by statistical analysis. Weighting the data by the 95% confidence limits of the respective rotations and fitting with first-, second-, and third-order polynomials, we find that a third-order polynomial (inset in Fig. 3) fits the data significantly better than a first-order polynomial (95% confidence) or second-order curve (50% confidence). In this analysis, an F test was used on chi-square statistics, with the greater number

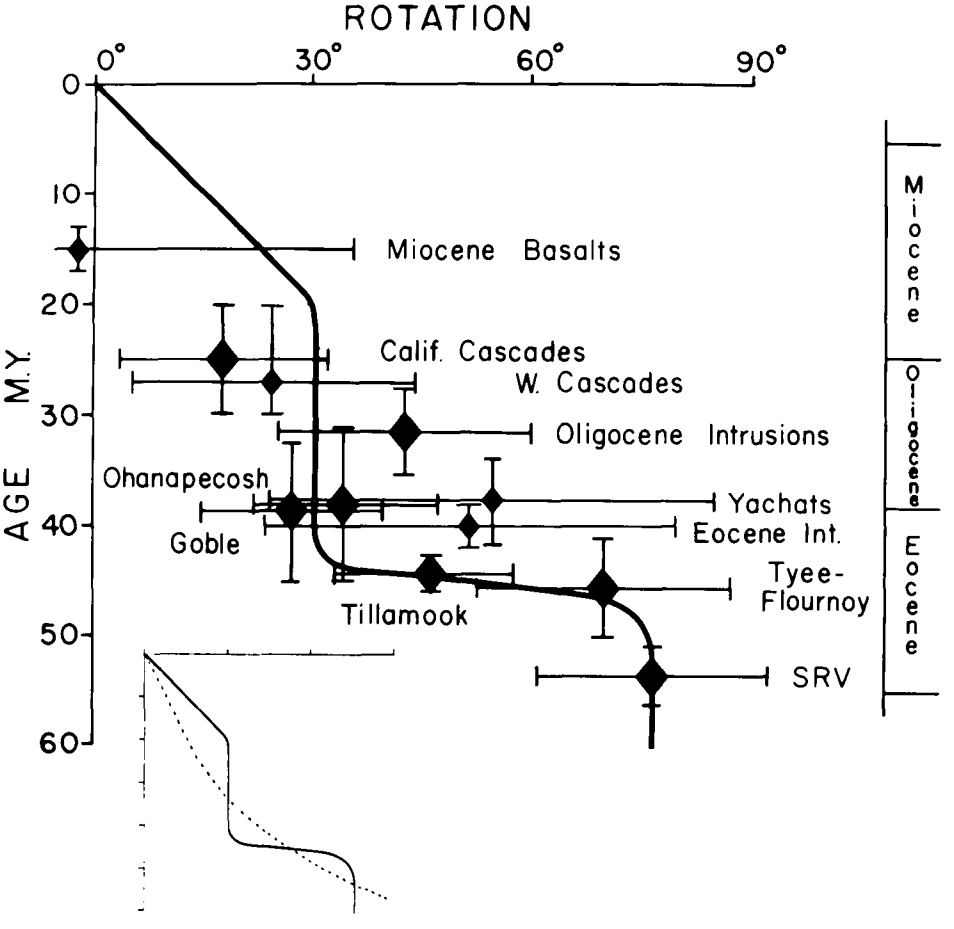


Figure 3. Rotation versus age for geologic units in Oregon Coast Range and Cascades. Rotation error bars are 95% confidence limits and were computed as noted in Table 1. Inset shows weighted least-squares fit of data to third-order polynomial curve (dotted) superimposed on our suggested curve (solid), the shape of which is based in part upon our interpretation of regional geology. Data point for Oligocene intrusions is average of data from Beck and Plumley (1980) and Clark (1969) collected at sampling sites OI and MP of Figure 1.

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of degrees of freedom in the higher order polynomials taken into account. Although this analysis does not prove that the third-order polynomial fit is correct, it suggests that the best fit curve may be more complicated than a simple straight line corresponding to rotation at a constant rate.

PALEOGEOGRAPHIC RECONSTRUCTIONS

Magill and others (1981) recently proposed that the regional geology and paleomagnetic data can best be explained by a model with two episodes of rotation: Phase I rotation during middle Eocene time and Phase II rotation beginning in the Miocene. Our present results from the Cascades support such a two-phase rotation. The Coast Range must have rotated with the Cascades about 25° clockwise during the past 25 m.y. However, lower and middle Eocene rocks of the Coast Range are rotated as much as 75°, so it appears that about 50° of Coast Range rotation occurred prior to 25 m.y. B.P. Our interpretation, based mainly on geologic considerations, is that this earlier Phase I was an accretionary rotation during the middle Eocene and was followed by a period of tectonic quiescence during the late Eocene and Oligocene which in turn was followed by Phase II rotation. A rotational history for the Coast Range and Cascades corresponding to this two-phase model is shown by the heavy curve in Figure 3. The curve fits the paleomagnetic data as well as the third-order polynomial and better than a straight line.

Phase I of the rotation appears to have occurred during middle Eocene time and to have resulted in about 46° of clockwise rotation (Magill and others, 1981). During the Eocene the Coast Range existed as a seamount province (Snavely and others, 1968; Dickinson, 1976), associated with submarine and subaerial basaltic volcanism, and was covered by turbidites (Snavely and others, 1964; Lovell, 1969) as the seamounts neared the North American continent. The Eocene clockwise rotation of the Coast Range seamount province may have occurred when it became trapped between two active trenches. As the wedge of oceanic crust between the trenches was partially subducted, the crust and overlying seamounts rotated clockwise toward the eastern trench. The geologic basis for this model was discussed by Magill and others (1981). The present-day rotation of the Philippine Sea plate, bounded by the Izu-Bonin and Ryukyu Trenches, may be a modern analogue of Phase I rotation.

In the Phase II rotation of our model, the Cascades, Coast Range, and Klamaths rotated together as one block (Fig. 4), accompanied by an extension of the Basin and Range province (see Stewart, 1978) similar to that proposed by Eaton (1979). Phase II began about 20 m.y. B.P. and resulted in about 27° of clockwise rotation. During this rotation the Cascades, Coast Range, Klamaths, and northern Sierra Nevada all moved westward together, retaining their east-west spacing. The distribution of late Cenozoic strike-slip and extensional faults in the western United States supports the existence of an extensional-strain maximum in northern Nevada that decreases rapidly to the north (Fig. 1). Such a distribution of extension would produce the observed clockwise rotation of all seaward terranes.

In support of the concept that the major mountain ranges moved as coherent blocks, with little internal deformation, Hamilton and Myers (1966) and Hamilton (1978) noted the absence of folding and faulting in the Sierra Nevada and Idaho batholiths and proposed that the batholiths behave tectonically as small crustal plates. We note that the extensional faults common in the western United States not only are absent from the Sierra Nevada but also are sparse within the Cascades, Oregon Coast Range, and Klamaths (Fig. 1). Similarly, the high heat flow of the Basin and Range province and Cascades is not present in the main mountain blocks. Lachenbruch and Sass (1978) have proposed that the high heat flow of the Basin and Range is related to convective heat transport associated with extensional intrusion of magma into a thinned crust. The absence of high heat flow and extensional faults and folding in the mountain blocks suggests that to first order, the extensional deformation in the western United States has been confined to the Basin and Range province and is absent in the Sierra Nevada, Coast Range, Klamaths, and Cascades. Together with the consistency of the paleomagnetic data, these observations provide the basis for our treating the Cascades, Coast Range, Klamaths, and Sierra Nevada as coherent blocks in our reconstructions.

During Basin and Range extension, as these blocks moved westward and rotated, the Klamath–Sierra Nevada junction behaved as a hinge point, which accommodated differential rotation between the Sierra Nevada block and the Klamath–Coast Range block. In this rigid-plate model, the magnitude of extensional strain in the Basin and Range is of the order of 80%. This amount of extension, although high, is still within the range of values proposed by Hamilton and Myers (1966), Hamilton (1978), and Elston (1978).

The reconstructed positions of the major crustal blocks at 20 m.y. B.P. (Fig. 4) were constrained in part by the paleomagnetic data and in part by geologic considerations. The orientations of the Coast Range, Cascade, and Klamath blocks are consistent with the paleomagnetic data previously discussed. In addition, somewhat scattered paleomagnetic data from the Sierra Nevada indicate a mean counterclockwise rotation of 4°, with large confidence circles (see Magill and Cox, 1980) that limit the amount of counterclockwise rotation of the Sierra Nevada to about 15°. We have also used the following two geologic constraints: (1) Major east-west motion between the Klamath Mountains and the Sierra Nevada is precluded by the occurrence of Upper Cretaceous marine sediments in depositional contact with the rocks of these two blocks. In our reconstruction, the east-west distance between the Sierra Nevada and Klamath blocks has been kept constant. (2) The most easterly possible position for the southern Sierra Nevada was determined by closing the Cenozoic grabens in the

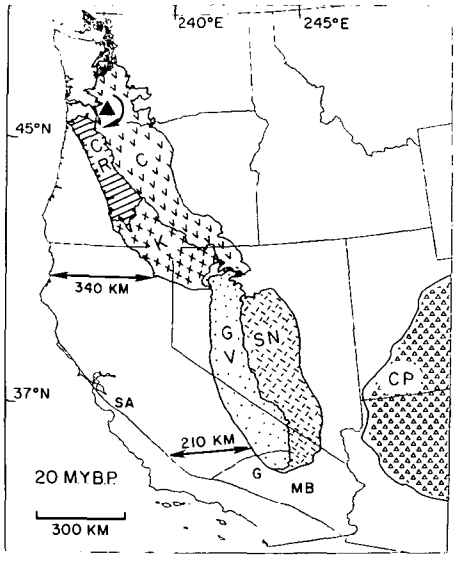


Figure 4. Position of primary tectonic blocks at beginning of Phase II rotation, 20 m.y. B.P. Triangle = rotation pivot at northern end of Coast Range block. Arrows indicate magnitudes of Basin and Range extension that took place east of primary blocks. Reference frame for reconstructions is provided by present-day state boundaries, San Andreas fault (SA), Garlock fault (G), and Mojave block (MB), all shown in their present-day positions. CR = Oregon Coast Range, C = Cascade Range, K = Klamath Mountains, SN = Sierra Nevada, GV = Great Valley, CP = Colorado Plateau.

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Basin and Range province against the Colorado Plateau. The parameters of the rotations used in constructing Figure 4 are given in Magill and Cox (1980) and Magill and others (1981), together with a fuller discussion of the model in the context of the regional geology.

The displacement of the Coast Range, Cascades, and Klamaths from their 20 m.y. B.P. positions to their present position can be described by the finite rotation pole shown as a triangle in Figure 4. If, as seems likely, the movement of the Coast Range was not along a perfect arc, then the instantaneous rotation poles during the rotational motion may have been displaced from this pole. However, rotation about poles in the general location shown would preserve continuity of the Cascades and would produce moderate compression and possibly uplift in the Cascades of southern Washington in accord with the Miocene geology of this region (Hammond, 1979). Our pole is also consistent with opening of the Basin and Range along a southwest-northeast trend, in agreement with the orientation of early rifts and extensional basins (Eaton, 1979; Zoback and Thompson, 1978). Using independent tectonic data from the Basin and Range province, Eaton (1979) used a pole located several hundred kilometres north of ours to describe the early extension of the Basin and Range province. The similarity of the two poles suggests that the tectonics in the two regions are driven by a common underlying process. The small but significant difference between the poles suggests that the tectonic responses of the two regions may not be identical, possibly because of differences in local factors such as boundary conditions imposed by the Sierra Nevada, Colorado Plateau, and northern Cascade region.

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